

BARAA JADAAN

AI Engineer | Full-Stack Developer | RAG Systems & MLOps

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SUMMARY

AI Engineer who ships LLM systems end to end — retrieval-augmented generation, agentic tool-calling pipelines, and QLoRA fine-tuning — from data pipeline through Dockerized cloud deployment. Bachelor's degree in Artificial Intelligence & Information Technology paired with roughly three years of full-stack and end-to-end software engineering, giving equal command of building ML/NLP models and the MLOps needed to run them reliably in production.

EDUCATION

Bachelor of Engineering in Artificial Intelligence and Information Technology

Damascus University

TECHNICAL SKILLS

LLM & Generative AI: RAG architecture, agentic tool-calling (ReAct-style orchestration), prompt engineering, LoRA/QLoRA fine-tuning (Unsloth, PEFT, TRL, bitsandbytes), GGUF quantization, Qwen & Claude model families

Machine Learning & NLP: scikit-learn, TensorFlow, PyTorch, Transformers, hybrid retrieval (dense + BM25), embedding models (Voyage, Qwen-Embedding), evaluation & experiment design

MLOps & Deployment: Docker, GitHub Actions CI/CD, MLflow, FastAPI, LanceDB/ChromaDB, Oracle Cloud (ARM)

Programming & Data: Python, Pandas, NumPy, SQL, JavaScript/TypeScript

Full-Stack & Mobile: React.js, React Native, Flutter, .NET, REST & SSE APIs, Supabase, Redux/Zustand

Languages: Arabic (Native), English (B2, TOEFL)

FEATURED AI/ML PROJECTS

Arabic Poetry Semantic Search & RAG Agent (Python, FastAPI, LanceDB, Docker, GitHub Actions, MLflow)

- **Retrieval engine:** Engineered a hybrid RAG pipeline over a 355k+ classical Arabic verse corpus — combining LanceDB dense vector search, Tantivy/BM25 full-text search, and Reciprocal Rank Fusion (RRF), indexed with IVF-PQ for sub-200ms latency on memory-constrained hardware.
- **Embedding architecture:** Benchmarked a local Voyage GGUF model, diagnosed a 0% semantic recall failure using a 25-query golden evaluation harness, and re-architected with Qwen3-Embedding (1024-dim Matryoshka truncation + Float16) — boosting Recall@1 from 0% to 94.4% (100% Recall@5).
- **Agentic orchestration:** Implemented a zero-framework ReAct agent in Python featuring a custom 3-state SSE stream splitter (routing <think> reasoning tokens to a live UI drawer) and fuzzy/regex fallback parsers to recover from LLM tool-calling syntax anomalies.
- **Production Architecture:** Shipped a dual-engine architecture (local GGUF offline mode vs. cloud API mode) with multi-key automatic failover for HTTP 429 quota resilience; decoupled the frontend (GitHub Pages) from a containerized FastAPI backend with strict CORS controls.
- **MLOps:** Instrumented MLflow tracking for retrieval metrics (Recall@k, MRR, latency), isolated dependencies into modular uv groups, and established an automated GitHub Actions CI pipeline executing deterministic regression test fixtures on every push.

Arabic Poetry Meter Classification — LLM Fine-Tuning (QLoRA, Qwen2.5-7B, Unsloth, TRL, PEFT, GGUF)

- **Fine-tuning:** Attached a rank-16 QLoRA adapter to all attention and MLP projection layers of Qwen2.5-7B-Instruct via Unsloth (4-bit), training 40.4M of the model's 7.6B parameters (0.53%) over 3 epochs on 5,000 balanced classical Arabic verses across 5 poetic meters (■■■■■).
- **Diagnosis-driven optimization:** Resolved base-model majority bias (guessing Tawil 84% of the time) and prevented training model collapse by implementing response-only cross-entropy loss masking to isolate gradient updates to the target meter labels.
- **Results:** Benchmarked on a 250-verse held-out test set (50/meter), achieving +32.0 point accuracy jump over the base model with 0% invalid formatting, evaluated via a custom statistical benchmarking harness.
- **Deployment & serving:** Eliminated train-serve distribution skew by standardizing inference prompt templates, then merged adapter weights and exported to 4-bit GGUF (q4_k_m) for lightweight local CPU inference via llama-cpp-python.

PROFESSIONAL EXPERIENCE

End-to-End Software Engineer — Rafeek Darbak (Feb 2026 – Present)

- Own the Flutter-based ride-sharing app “**Rafiq Darbak**” (████████████████) end to end — built and shipped across iOS and Android. iOS deployment via Codemagic/App Store Connect and Android release management via Google Play Console including test releases.
- Built, maintained and resolved production issues of the platform's admin panel, iOS and Android.
- Shipped major features including voice messaging, real-time messaging, live location sharing, radius-based matching, and the Explore screen, backed by Supabase (PostgreSQL) for data storage, authentication, and Row-Level Security policies.

Frontend Developer — TabTabGo (Oct 2024 – Feb 2026)

- Sole frontend developer on a Customer & Network Management Suite for a Danish ISP — built the admin portal, web onboarding portal, and a companion React Native mobile app, including direct client calls to gather and refine requirements.
- Contributed frontend and dashboard development to a multi-platform Social Monitoring & Dashboard product — a listening and analytics engine tracking talent and campaign performance across social platforms.
- Took on select backend tasks in C#/NET during later months at the job, extending past the primary frontend scope.

Frontend Developer — ATC Systematic (Rafeed) (Dec 2023 – Aug 2024)

- Built the frontend for Lumytic, ATC Systematic's ERP platform for the lighting industry, including the public marketing website, CMS, and CRM systems.

ADDITIONAL PROJECTS

- **Arabic Named Entity Recognition:** Sequence labeling for Arabic text using statistical, machine-learning, and deep-learning approaches.
- **Grammar Correction with Transformers:** Fine-tuned a pre-trained transformer model to correct grammar in user-submitted text.
- **Medical Q&A; Data Analysis:** Cleaned, analyzed, and classified thousands of real-world questions from an online medical-advice platform.
- **Exam Proctoring System (Graduation Project):** Head-pose estimation, face recognition, eye tracking, and object detection to flag cheating in online exams.
- **Sound Classification for Hearing-Impaired Parents:** Audio classification for an assistive alerting system.
- **Front-End Applications:** E-commerce platform, gaming discovery portal, real-time stock tracker, and movie search app.